

Section 6E. PCR for $M = 1$

Victor M. Preciado, PhD MIT EECS
Dept of Electrical & Systems Engineering
University of Pennsylvania
preciado@seas.upenn.edu

PCR Criterion: $M = 1$

Variance of Linear Combinations:

- ▶ Assume a vector of r.v.'s $X = [X_1, X_2, \dots, X_p]^T$ and coefficients $\Phi_1^T = [\phi_{11}, \phi_{12}, \dots, \phi_{1p}]$.
- ▶ What is the variance of $Z_1 = \phi_{11}X_1 + \phi_{12}X_2 + \dots + \phi_{1p}X_p = \Phi_1^T X$? For simplicity, let us assume $\mathbb{E}[X] = \mathbf{0}$; hence,

$$\begin{aligned}\text{Var}[Z_1] &= \text{Var}[\Phi_1^T X] = \mathbb{E}[(\Phi_1^T X)^2] \\ &= \mathbb{E}[\Phi_1^T X X^T \Phi_1] = \Phi_1^T \mathbb{E}[X X^T] \Phi_1 \\ &= \Phi_1^T \Sigma_X \Phi_1\end{aligned}$$

PCR Criterion: $M = 1$ (cont.)

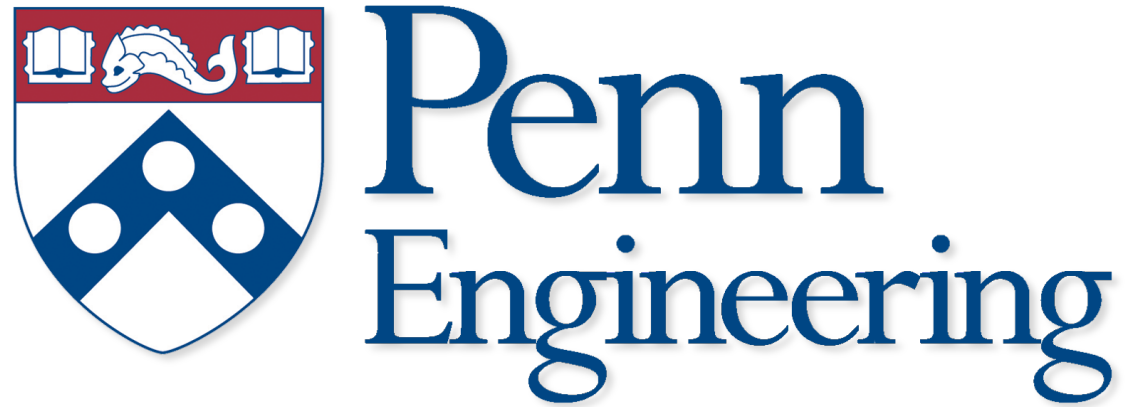
PCR Problem for $M = 1$: Find an input variable $Z_1 = \Phi_1^T X$ such that $\text{Var}[Z_1]$ is maximized

- ▶ We have that $\text{Var}[Z_1] = \Phi_1^T \Sigma_X \Phi_1$; hence, to find the vector of coefficients Φ_1 maximizing the variance, we need to solve the following optimization problem:

$$\begin{aligned} \max_{\Phi_1 \in \mathbb{R}^p} \quad & \Phi_1^T \Sigma_X \Phi_1 \\ \text{s.t.} \quad & \|\Phi_1\| = 1 \end{aligned}$$

where the constraint $\|\Phi_1\| = 1$ is included to avoid the variance blowing up to infinity

- ▶ Using linear algebra, the solution to the above optimization problem is (proof not included) $\Phi_1 = \mathbf{v}_1$, where $\mathbf{v}_1$ is the unit-length eigenvector of Σ_X corresponding to its largest eigenvalue λ_1



Copyright 2020 University of Pennsylvania
No reproduction or distribution without permission.